A finite-dimensional exact witness for the separable transfinite SQ example

Run `fa_banach_001`, lane 10

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Source and target

The source paper is A. Avilés, S. Ciaci, J. Langemets, A. Lissitsin, and A. Rueda Zoca, *Transfinite almost square Banach spaces*, arXiv:2204.13449. In Definition 2.1 the paper defines $\mathrm{SQ}_{<\kappa}$: for every $A \subset S_X$ with $|A| < \kappa$, there is $y \in S_X$ such that

$$\|x \pm y\| \le 1 \qquad (x \in A).$$

For uncountable κ, Proposition 2.2 gives an equivalent formulation in terms of subspaces of density $< \kappa$. The paper explicitly notes that the corresponding $\kappa = \aleph_0$ version is unclear. Shortly afterward it constructs a separable $\mathrm{SQ}_{<\aleph_0}$ example as a c_0-sum of odd-function spaces.

This packet proves that this concrete separable example has the stronger finite-dimensional exact witness property.

Definitions

For $m \ge 1$, let

$$E_m = \{f \in C(S_{\mathbb{R}^m}) : f(-s) = -f(s) \text{ for all } s \in S_{\mathbb{R}^m}\},$$

with the supremum norm. Let

$$X = \Big(\bigoplus_{m=1}^{\infty} E_m\Big)_{c_0}.$$

Thus an element $x \in X$ is a sequence $x = (x_m)$, $x_m \in E_m$, with $\|x_m\|_\infty \to 0$, and $\|x\| = \sup_m \|x_m\|_\infty$.

Proof Intuition

The finite-set proof in the source works by moving to one high-dimensional odd function coordinate and using Borsuk-Ulam to find a point where the relevant odd functions all vanish. For a finite-dimensional subspace F, one should not choose a coordinate merely matching the number of listed vectors; instead choose $m > \dim F$. Then the whole coordinate shadow $P_m F$ has dimension strictly less than m, so all its functions vanish at a common point of $S_{\mathbb{R}^m}$.

At that coordinate, the function

$$q(s) = \sup\{|(P_m z)(s)| : \ z \in F, \ \|z\| \le 1\}$$

measures how much of the unit ball of F is visible at s. Since $q(s_0) = 0$ at the Borsuk-Ulam point, the odd function $(1-q)h$, where $h(s) = \langle s, s_0 \rangle$, has norm one and leaves exactly enough pointwise room for every vector in F.

Theorem 1. *Let $X = (\bigoplus_{m=1}^{\infty} E_m)_{c_0}$ be the separable example above. For every finite-dimensional subspace $F \subset X$, there exists $u \in S_X$ such that*

$$\|z + tu\| \le \max\{\|z\|, |t|\} \qquad (z \in F,\ t \in \mathbb{R}).$$

Consequently X is $\mathrm{SQ}_{<\aleph_0}$, and in fact satisfies the finite-dimensional exact formulation for $\kappa = \aleph_0$.

Proof. Let $F \subset X$ be finite-dimensional and put $d = \dim F$. Choose an integer $m > d$. Let $P_m : X \to E_m$ be the m-th coordinate projection. The subspace $P_m F \subset E_m$ has dimension $r \le d < m$. Choose a basis $f_1, \dots, f_r$ of $P_m F$. The map

$$\Phi : S_{\mathbb{R}^m} \to \mathbb{R}^r, \qquad \Phi(s) = (f_1(s), \dots, f_r(s))$$

is continuous and odd. By the Borsuk-Ulam theorem, since $r < m$, there is $s_0 \in S_{\mathbb{R}^m}$ such that $\Phi(s_0) = 0$. Hence

$$f(s_0) = 0 \qquad (f \in P_m F).$$

Define

$$q(s) = \sup\{|(P_m z)(s)| :\ z \in F,\ \|z\| \le 1\} \qquad (s \in S_{\mathbb{R}^m}).$$

Because the unit ball of F is compact and the evaluation map is continuous, q is continuous. Also q is even, $0 \le q \le 1$, and $q(s_0) = 0$. Let $h(s) = \langle s, s_0 \rangle$, an odd norm-one function on $S_{\mathbb{R}^m}$, and set

$$g(s) = (1 - q(s))h(s).$$

Then $g \in E_m$, $\|g\|_\infty = 1$, since $g(s_0) = 1$, and $|g(s)| \le 1 - q(s)$ for all s.

Let $u \in X$ be zero in every coordinate except the m-th coordinate, where $u_m = g$. Then $u \in S_X$. Fix $z \in F$, $t \in \mathbb{R}$, and write $M = \max\{\|z\|, |t|\}$. For every coordinate $j \ne m$,

$$\|(z + tu)_j\|_\infty = \|z_j\|_\infty \le \|z\| \le M.$$

For the m-th coordinate, if $M = 0$ there is nothing to prove. Otherwise, for every $s \in S_{\mathbb{R}^m}$,

$$\begin{aligned} |z_m(s) + tg(s)| &\le \|z\| \sup_{\|w\| \le 1,\, w \in F} |(P_m w)(s)| + |t|\,|g(s)| \\ &\le \|z\|\ q(s) + |t|\,(1 - q(s)) \\ &\le M. \end{aligned}$$

Taking the supremum over s and then over all coordinates gives $\|z + tu\| \le M$, as required.

For a finite set $A \subset S_X$, apply the theorem to $F = \mathrm{span}(A)$ and $t = \pm 1$. This yields $\|x \pm u\| \le 1$ for every $x \in A$, so X is $\mathrm{SQ}_{<\aleph_0}$. □

Verification and limitations

The only external ingredient is the standard Borsuk-Ulam theorem: every continuous odd map $S^{m-1} \to \mathbb{R}^r$ has a zero when $r < m$. The proof is otherwise a direct pointwise estimate in the source's c_0-sum example.

This is a partial result. It does not prove that every $\mathrm{SQ}_{<\aleph_0}$ space satisfies the finite-dimensional exact formulation. It also does not answer the source question of whether c_0 admits an equivalent $\mathrm{SQ}_{<\aleph_0}$ renorming, nor the general $c_0(\kappa)$-renorming question.

Novelty check: the run indexes contained only the prior attempt

`2204.13449_sq_kappa_renorming_density_barrier.md`.

A bounded web search on 2026-06-18 for the source title together with $\mathrm{SQ}_{<\kappa}$, $c_0(\kappa)$, and $\mathrm{SQ}_{<\aleph_0}$ found the source paper but no later exact treatment of this finite-dimensional witness strengthening.

References

References

[1] A. Avilés, S. Ciaci, J. Langemets, A. Lissitsin, and A. Rueda Zoca, *Transfinite almost square Banach spaces*, arXiv:2204.13449.

[2] K. Borsuk, *Drei Sätze über die n-dimensionale euklidische Sphäre*, Fund. Math. 20 (1933), 177–190.